

**UNIVERSITY OF ENGINEERING AND TECHNOLOGY
PESHAWAR (NOWSHERA CAMPUS)**

COMPUTER PROGRAMMING

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<i>Subject:</i>	<i>Computer Programming</i>
<i>Department:</i>	<i>Electrical Engineering</i>
<i>Assignment:</i>	<i>Two (02)</i>

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Program No: 1

```
#include <iostream>
using namespace std;
int main() {
    int arr[10], max, min, sum = 0;

    // Taking input from user
    cout << "Enter 10 integers:\n";
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
```

```
}
```

```
// Initialize max and min with first  
element max = min = arr[0];
```

```
// Loop to find max, min and sum  
for(int i = 0; i < 10; i++) {  
    if(arr[i] > max) max = arr[i]; // Update  
    max if(arr[i] < min) min = arr[i]; //  
    Update min sum += arr[i]; // Add to  
    sum  
}
```

```
// Display results  
cout << "Maximum: " << max << endl;  
cout << "Minimum: " << min << endl;  
cout << "Average: " << sum / 10.0 <<  
endl;
```

```
return 0;  
}
```

Program No: 2

```
#include <iostream>  
using namespace std;
```

```
// Function to reverse array  
void reverseArray(int arr[], int size)  
{ for(int i = 0; i < size/2; i++) {  
    swap(arr[i], arr[size - i - 1]); // Swap  
elements }  
}
```

```
int main() {  
    int arr[5] = {1,2,3,4,5};  
    // Display original array  
    cout << "Before Reversal: ";  
    for(int i=0;i<5;i++) cout << arr[i] << " ";
```

```
reverseArray(arr, 5); // Function call
```

```
// Display reversed array
```

```
cout << "\nAfter Reversal: ";
```

```
for(int i=0;i<5;i++) cout << arr[i] << " ";
```

```
return 0;
```

```
}
```

Program No: 3

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
int A[3][3], B[3][3], C[3][3];
```

```
// Input Matrix A
```

```
cout << "Enter Matrix A:\n";
```

```
for(int i=0;i<3;i++)
```

```
for(int j=0;j<3;j++)
```

```
cin >> A[i][j];
```

```
// Input Matrix B
```

```
cout << "Enter Matrix B:\n";
```

```
for(int i=0;i<3;i++)
```

```
for(int j=0;j<3;j++)
```

```
cin >> B[i][j];
```

```
// Add matrices
```

```
for(int i=0;i<3;i++)
```

```
for(int j=0;j<3;j++)
```

```
C[i][j] = A[i][j] + B[i][j];
```

```
// Display result
```

```
cout << "Result Matrix:\n";
```

```
for(int i=0;i<3;i++) {
```

```
for(int j=0;j<3;j++)
```

```
cout << C[i][j] << " ";
```

```
cout << endl;
```

```
}
```

```
return 0;
}
```

Program No: 4

```
#include <iostream>
using namespace std;
```

```
int main()
{ string
names[5];
```

// Input names

```
cout << "Enter 5 names:\n";
for(int i=0;i<5;i++)
cin >> names[i];
```

// Sorting using simple

```
comparison for(int i=0;i<5;i++)
{ for(int j=i+1;j<5;j++)
{ if(names[i] > names[j]) {
swap(names[i], names[j]); // Swap
names }
}
}
```

// Display sorted names

```
cout << "Sorted Names:\n";
for(int i=0;i<5;i++)
cout << names[i] << endl;
```

```
return 0;
}
```

Program No: 5

```
#include <iostream>
using namespace std;
```

// Function to calculate total load

```
double totalLoad(double arr[], int
```

```

size)
{ double sum = 0;
for(int i=0;i<size;i++) sum += arr[i]; re
turn sum;
}

```

// Function to find peak load

```

double peakLoad(double arr[], int size)
{ double max = arr[0];
for(int i=1;i<size;i++)
if(arr[i] > max) max = arr[i];
return max;
}

```

```

int main() { dou
ble load[24];

```

// Input load data

```

cout << "Enter 24-hour load:\n";
for(int i=0;i<24;i++)
cin >> load[i];

```

// Display results

```

cout << "Total Load: " << totalLoad(load,24) <<
endl; cout << "Peak Load: " <<
peakLoad(load,24) << endl;

```

```

return 0;
}

```

Program No: 6

```

#include <iostream>
using namespace std;

```

// Function to find topper

```

void topper(string names[], int marks[], int
size) { int max = marks[0], index = 0;
for(int i=1;i<size;i++)
{ if(marks[i] > max)
{ max = marks[i];

```

```

index = i;
}
}
cout << "Topper: " << names[index] <<
endl; }

```

// Function to display pass/fail

```

void passFail(string names[], int marks[], int
size) { for(int i=0;i<size;i++) {
cout << names[i] << " - ";
if(marks[i] >= 50) cout << "Pass\n";
else cout << "Fail\n";
}
}

```

// Function to search student

```

void search(string names[], int size, string
key) { for(int i=0;i<size;i++) {
if(names[i] == key) {
cout << "Found: " << names[i] <<
endl; re turn;
}
}
cout << "Not Found\n";
}

```

```

int main() {
string names[3] =
{"Ali","Ahmed","Sara"}; int marks[3] =
{80,45,90};

```

```

topper(names, marks, 3);
passFail(names, marks, 3);
search(names, 3, "Ahmed");

```

```

return 0;
}

```

Program No: 7

```

#include <iostream>
using namespace std;
// Calculate total energy
double totalEnergy(double arr[]) { double sum=0;
for(int i=0;i<24;i++) sum+=arr[i];
return sum;
}

// Find hour of maximum generation
int maxHour(double arr[])
{ int index=0;
for(int i=1;i<24;i++)
if(arr[i] > arr[index]) index = i; return index;
}

int main() {
double energy[24];

// Input data
for(int i=0;i<24;i++)
cin >> energy[i];

cout << "Total Energy: " << totalEnergy(energy) << endl;
cout << "Max Hour: " << maxHour(energy) << endl;
cout << "Average: " << totalEnergy(energy)/24 << endl;

return 0;
}

```

Program No: 8

```

#include <iostream>
using namespace std;

int main()
{ double
v[20];
int low=0, high=0;

```

// Input voltage readings

```
for(int i=0;i<20;i++) cin >> v[i];
```

// Check safe limits

```
for(int i=0;i<20;i++)  
{ if(v[i] < 3.0) low++;  
  if(v[i] > 4.2) high++;  
}
```

// Find max and min voltage

```
double max=v[0], min=v[0];  
for(int i=1;i<20;i++) {  
  if(v[i] > max) max=v[i];  
  if(v[i] < min) min=v[i];  
}
```

```
cout << "Low: " << low <<  
endl; cout << "High: " <<  
high << endl; cout << "Max: "  
<< max << endl; cout <<  
"Min: " << min << endl;
```

// Warning if unsafe

```
if(low>0 || high>0)  
cout << "Warning: Unsafe readings!\n";
```

```
return 0;  
}
```

Program No: 9

```
#include <iostream>  
using namespace std;
```

```
int main()  
{ int arr[4][7];
```

// Input data

```
for(int i=0;i<4;i++)  
for(int j=0;j<7;j++)  
cin >> arr[i][j];
```


// Calculate total per region

```
for(int i=0;i<4;i++)
{ int sum=0;
for(int j=0;j<7;j++)
sum+=arr[i][j];
cout << "Region " << i+1 << " Total: " << sum
<< endl; }
return 0;
}
```

Program No: 10

```
#include <iostream>
using namespace std;
```

```
int main()
{ int att[5][7];
```

// Input attendance

```
for(int i=0;i<5;i++)
for(int j=0;j<7;j++)
cin >> att[i][j];
```

```
int max=0, index=0;
```

// Calculate attendance

```
for(int i=0;i<5;i++)
{ int sum=0;
for(int j=0;j<7;j++)
sum+=att[i][j];
```

```
cout << "Student " << i+1 << ": " << sum << endl;
```

```
if(sum > max) {
max = sum;
index = i;
}
}
```

```
cout << "Highest Attendance: Student " << index+1 <<
```

```
endl; re turn 0;  
}
```